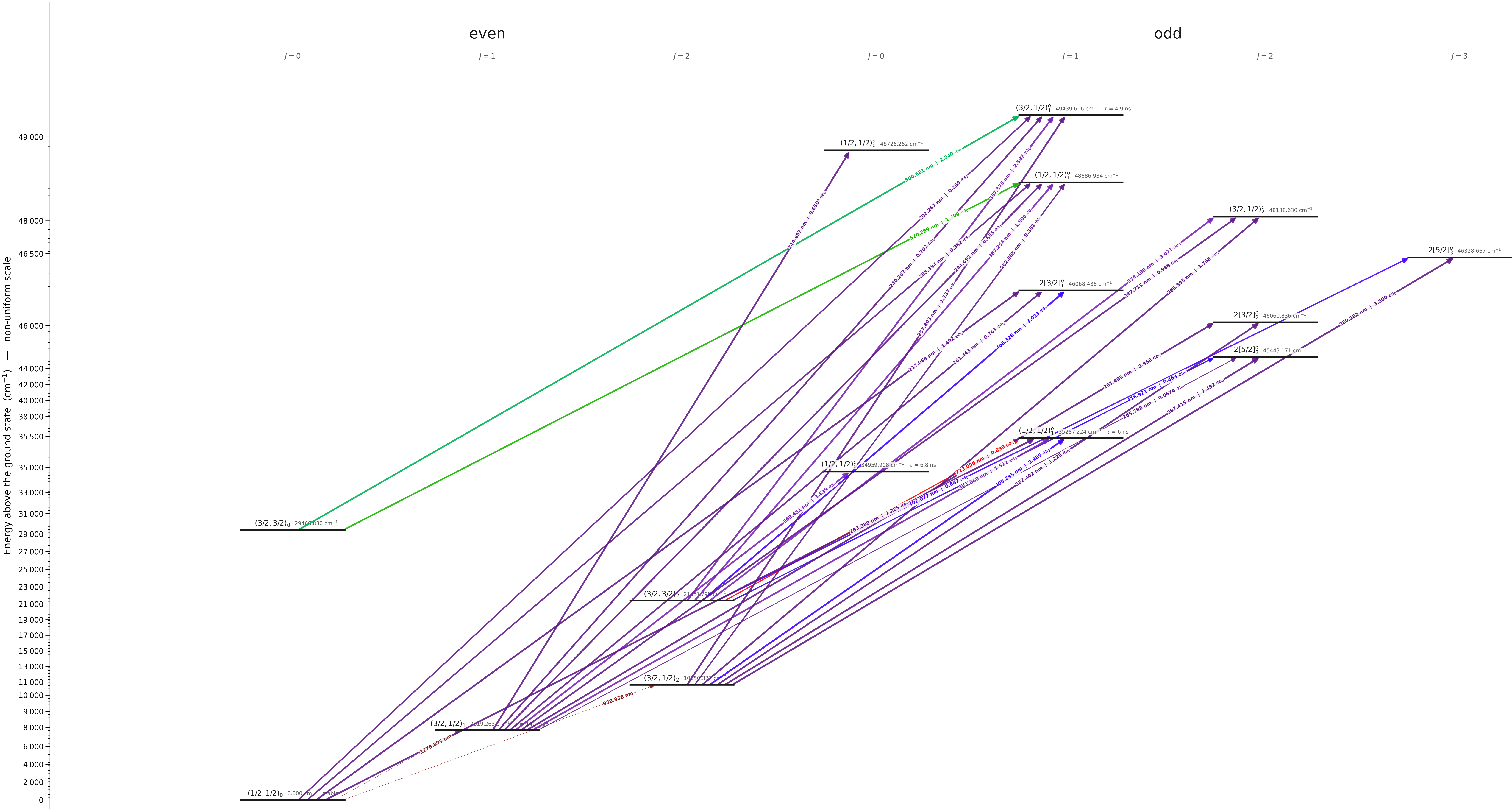
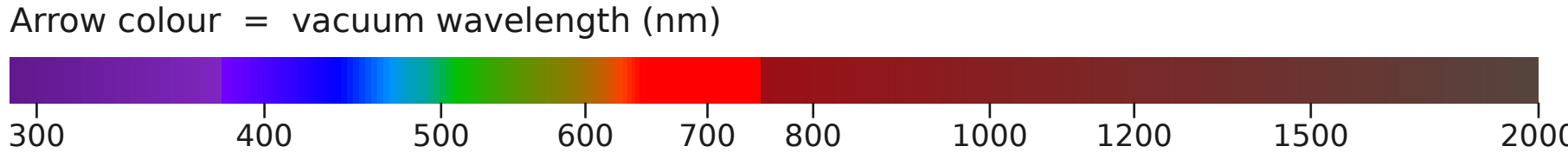


# Lead

Pb (Pb I). Strongest classified transitions below 2 μm (NIST): 15 levels, 31 lines.

Pb + ionisation limit 59819.56 cm<sup>-1</sup> = 7.41668 eV (167.169 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $[J||e\mathbf{r}||J']$  in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                                                     | λ vac (nm) | λ air (nm) | ν (THz)   | $[J  e\mathbf{r}  J']$ (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|----------------------------------------------------------------|------------|------------|-----------|-------------------------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| (1/2, 1/2) <sub>0</sub> – (1/2, 1/2) <sub>1</sub> <sup>o</sup> | 283.3887   | 283.3053   | 1057.8844 | 1.285                                     | 4.900e+07            | 26.53          | 29.4       | ground-state resonance line                                  |
| (3/2, 1/2) <sub>1</sub> – (1/2, 1/2) <sub>1</sub> <sup>o</sup> | 364.0605   | 363.9568   | 823.4688  | 1.512                                     | 3.200e+07            | 26.53          | 19.2       | isotope shifts and hyperfine structure measured by Dopple... |
| (3/2, 1/2) <sub>1</sub> – (1/2, 1/2) <sub>0</sub> <sup>o</sup> | 368.4511   | 368.3462   | 813.6561  | 1.839                                     | 1.370e+08            | 23.41          | 100        | proposed laser-cooling / magneto-optical-trap cycling line   |
| (3/2, 1/2) <sub>2</sub> – (1/2, 1/2) <sub>1</sub> <sup>o</sup> | 405.8953   | 405.7807   | 738.5956  | 2.985                                     | 9.000e+07            | 26.53          | 54         | the Pb I line used for lead abundances in stellar spectra    |
| (1/2, 1/2) <sub>0</sub> – (3/2, 1/2) <sub>2</sub>              | 938.9383   | 938.6807   | 319.2888  | -                                         | -                    | -              | -          | electric-quadrupole line measured by Faraday-rotation spe... |
| (1/2, 1/2) <sub>0</sub> – (3/2, 1/2) <sub>1</sub>              | 1278.8930  | 1278.5432  | 234.4156  | -                                         | 7.190e+00            | 1.137e-06      | 100        | magnetic-dipole line used for parity-nonconservation opti... |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## <sup>207</sup>Pb hyperfine structure (*I* = 1/2)

| Level                                | A (MHz)    | B (MHz) | Shift of each F level from the centre of gravity (MHz) |
|--------------------------------------|------------|---------|--------------------------------------------------------|
| (3/2, 1/2) <sub>1</sub>              | -2389.4(7) | -       | F=1/2: +2389.40 F=3/2: -1194.70                        |
| (3/2, 1/2) <sub>2</sub>              | 2600.8(9)  | -       | F=3/2: -3901.20 F=5/2: +2600.80                        |
| (3/2, 3/2) <sub>2</sub>              | 609.820(8) | -       | F=3/2: -914.73 F=5/2: +609.82                          |
| (1/2, 1/2) <sub>1</sub> <sup>o</sup> | 8802.0(16) | -       | F=1/2: -8802.00 F=3/2: +4401.00                        |
| 2[5/2] <sub>2</sub> <sup>o</sup>     | 3094(9)    | -       | F=3/2: -4641.00 F=5/2: +3094.00                        |
| 2[3/2] <sub>2</sub> <sup>o</sup>     | -1456.000  | -       | F=3/2: +2184.00 F=5/2: -1456.00                        |
| 2[3/2] <sub>1</sub> <sup>o</sup>     | -2884.000  | -       | F=1/2: +2884.00 F=3/2: -1442.00                        |
| 2[5/2] <sub>1</sub> <sup>o</sup>     | 2072(8)    | -       | F=5/2: -4144.00 F=7/2: +3108.00                        |
| (3/2, 1/2) <sub>2</sub> <sup>o</sup> | 1734.000   | -       | F=3/2: -2601.00 F=5/2: +1734.00                        |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition                                                     | λ vac (nm) | Isotopes | Shift (MHz) |
|----------------------------------------------------------------|------------|----------|-------------|
| (1/2, 1/2) <sub>0</sub> – (1/2, 1/2) <sub>1</sub> <sup>o</sup> | 283.389    | 204-208  | -4212.0(25) |
| (1/2, 1/2) <sub>0</sub> – (1/2, 1/2) <sub>1</sub> <sup>o</sup> | 283.389    | 206-208  | -2226.6(25) |
| (1/2, 1/2) <sub>0</sub> – (1/2, 1/2) <sub>1</sub> <sup>o</sup> | 283.389    | 207-208  | -1390.0(25) |
| (3/2, 1/2) <sub>2</sub> – (1/2, 1/2) <sub>1</sub> <sup>o</sup> | 405.895    | 206-208  | -2428.00    |
| (3/2, 1/2) <sub>2</sub> – (1/2, 1/2) <sub>1</sub> <sup>o</sup> | 405.895    | 204-208  | -4587.00    |
| (3/2, 1/2) <sub>2</sub> – (1/2, 1/2) <sub>1</sub> <sup>o</sup> | 405.895    | 207-208  | -1569.00    |
| (1/2, 1/2) <sub>0</sub> – (3/2, 1/2) <sub>1</sub>              | 1278.893   | 206-208  | 220.4(15)   |
| (1/2, 1/2) <sub>0</sub> – (3/2, 1/2) <sub>1</sub>              | 1278.893   | 207-208  | 138.0(17)   |

Shift = ν(first isotope) – ν(second isotope).